

Real-World Order of OperationsQuestion Bank - Additional 4.07 **ANSWER KEY**

*Show your work. Circle your final answer.***For questions 1 through 4, substitute and evaluate.**

1. $5x^2$ when $x = -4$

80

$$5(-4)^2 = 5(16)$$

2. $\frac{1}{2}mv^2$ when $m = 30$ and $v = 10$

1,500

$$15(100)$$

3. $\frac{a^2+b}{c}$ when $a = 8$, $b = 4$, $c = 17$

4*The bar groups the whole numerator: $64 + 4 = 68$.*

4. $3(x+y)^2$ when $x = -5$ and $y = 2$

27

$$3(-3)^2 = 3(9)$$

For questions 5 and 6, write and evaluate an expression. Give the unit.

5. A square room of side 13 feet is floored at 2.75 dollars a square foot, with a flat 90 dollars for delivery. Find the total cost.

$$13^2(2.75) + 90 = 554.75 \text{ dollars}$$

6. A rectangular patio 16 feet by 7 feet is sealed at 1.40 dollars a square foot, with a flat 35 dollars for equipment hire. Find the total cost.

$$16(7)(1.40) + 35 = 191.80 \text{ dollars}$$

The area is 112 square feet.

Two formulas.

7. Body mass index is $B = \frac{703w}{h^2}$, where w is a weight in pounds and h is a height in inches. Find B to the nearest tenth.

a. $w = 120$, $h = 62$

21.9

$$\frac{84,360}{3,844} = 21.945\dots$$

b. $w = 165$, $h = 68$

25.1

$$\frac{115,995}{4,624} = 25.085\dots$$

c. $w = 210$, $h = 74$

27.0

$$\frac{147,630}{5,476} = 26.959\dots$$

8. Kinetic energy is $K = \frac{1}{2}mv^2$, for mass m in kilograms and speed v in meters per second. Find K in joules.

a. $m = 1,500$, $v = 20$

300,000 joules

b. $m = 640$, $v = 9$

25,920 joules

c. $m = 55$, $v = 4$

440 joules

Working backwards.

9. Find the missing value in each situation.

a. $\frac{1}{2}mv^2 = 1,600$ and $v = 8$. Find m .

$m = 50$

$$32m = 1,600$$

b. $x^2 - 7 = 74$, where x is positive. Find x .

$x = 9$

- c. A square lawn of side n feet is turfed at 3.00 dollars a square foot with a flat 55 dollars delivery, and the total is 1,255 dollars. Find n .

$n = 20$

$$n^2(3.00) = 1,200 \text{ , so } n^2 = 400$$

Reading the situation.

10. Two problems use the numbers 8, 12 and 4. **A:** a printer costs 8 dollars to hire plus 12 dollars for each of 4 days. **B:** a printer costs 8 dollars for each of 12 reams, plus 4 dollars for delivery. Find both totals, and explain why circling the numbers would not have told you what to do.

A is $8 + 12(4) = 56$ dollars. B is $8(12) + 4 = 100$ dollars. The two problems carry the same three numbers, so the numbers cannot be what decides; what decides is which amount is repeated and how often.

Ask which number is a rate in each problem — that question has a different answer in A and in B, and no marking on the page can show it.

11. A square court of side 20 feet is resurfaced at 3.60 dollars a square foot, with a flat 200 dollars for the equipment. Which expression gives the total cost?

- ☐ A. $20^2(3.60 + 200)$
☐ B. $20(3.60) + 200$
☒ C. $20^2(3.60) + 200$
☐ D. $20^2 + 3.60 + 200$

C

12. Select ALL of the expressions whose value is 32 when $x = 4$.

- ☒ A. $x^2 + 16$
☒ B. $\frac{128}{x}$
☐ C. $2x \cdot 2$
☒ D. $x^2 \cdot 2$
☐ E. $(2x)^2$
☒ F. $2x^2$

A, B, D and F